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CHAPTER SEVEN

AI in Nigeria: Transforming Teaching and Learning: Challenges and Opportunities

Mary Oluwagbemisola Adetula

Department of Political Science,
Obafemi Awolowo University, Ile-Ife
adetulaoluwagbemisola@gmail.com
<https://orcid.org/0009-0001-5061-4137>

Background

Artificial intelligence (AI) is revolutionising various industries worldwide, including governance, finance, healthcare, and agriculture. This engineered or machine-based system can generate outputs such as predictions, recommendations, or decisions. AI systems operate with varying levels of autonomy. The growing global interest in AI highlights the urgent need for governance frameworks to address public concerns and ensure responsible research and deployment of this technology (Wodi, 2024). Organisations like the OECD and UNESCO are at the forefront of developing guidelines for ethical and trustworthy AI.

The OECD AI Principles promote inclusive growth, human rights, transparency, robustness, and international cooperation. These principles have influenced AI governance models such as the NIST AI Risk Management Framework (RMF) and the EU AI Act (Wodi, 2024). Countries including the United States, the United Kingdom, Canada, China, Singapore, and Nigeria have formulated their own AI strategies, often focusing on expanding AI research and development, fostering a skilled workforce, and creating a regulatory environment that encourages innovation while managing risks. Key ethical concerns, including bias, discrimination, privacy, and the impact of AI on employment and society, must be thoroughly addressed. Global initiatives like the US Blueprint for AI Bill of Rights and the Montreal Declaration for Responsible Development of Artificial Intelligence are working to tackle these issues (Wodi, 2024). AI holds significant promise in African nations for addressing unique challenges, particularly in financial inclusion and sustainable agriculture. To fully realise these benefits, providing equitable access to technology and skills is essential while narrowing the digital divide (Kishor, Patle, & Bhojane, 2023).

AI is transforming public services in Africa, including governance, policymaking, the economy, banking, agriculture, the judiciary, and education. However, the educational landscape faces significant challenges due to the widespread digital divide (Boateng, 2024). Limited computer access, internet connectivity, and stable electricity underscore more profound socio-economic disparities between countries (African Union, 2024). Nevertheless, Africa's youthful and tech-savvy population is eager to embrace innovation, presenting a unique opportunity for AI to bridge educational gaps and empower future generations (African Union, 2024). Increased smartphone use could enhance access to AI-powered educational tools, especially in low-resource contexts (Boateng, 2024; African Union, 2024).

Artificial Intelligence In Education: Insights From Various African Nations

An IoT and AI-based attendance tracking system in Rwanda has significantly reduced dropout rates in non-boarding secondary schools, as demonstrated at Wisdom School Musanze. This example provides valuable insights into how technology can positively impact school attendance and reduce dropout rates (Munyaneza et al., 2023). In Uganda, AI is showing potential for easing teaching workloads at public universities, offering a comprehensive view of AI's role in managing educational responsibilities (Namutebi, 2024).

In Tanzania, the use of ChatGPT in higher education institutions underscores AI's growing role in academic research, although the report also highlights ethical concerns regarding the integration of AI into research processes. However, Nkata (2020) notes an unsuccessful attempt to implement ICT as a teaching tool in Tanzanian secondary schools, primarily due to infrastructure limitations. In Egypt, AI integration has shown promising results in enhancing teaching and learning outcomes. There is the possibility of developing a virtual learning model based on educational data analysis and implementing another generative AI-based tutoring system for community schools in Upper Egypt (Saudi et al., 2023).

In South Africa, AI is proving to be a transformative force in higher education. ADvTech Group has led this movement, becoming the first company in the country to launch a large-scale AI-powered platform called ADvLEARN, which supports students in mathematics, physical science, and mathematical literacy. This platform allows learners to progress at their own pace, enhancing personalised learning experiences (Ntsobi & Bongani, 2024). Similarly, DreamBox Learning leverages adaptive learning technology to assist students in mathematics by aligning the content with each learner's skill level (Ntsobi & Bongani, 2024). Explainable AI models in South Africa also play a role in identifying students at risk of dropping out, enabling educators to provide targeted support that could reduce dropout rates (Bokgoshi et al., 2023). Additionally, AI contributes to curriculum evaluation by identifying learning gaps and personalising content, fostering improved educational outcomes (Pusporini & Nurdyanto, 2024). AI supports active learning, resource sharing, and virtual learning environments. However, the widespread adoption of AI in South African education faces several challenges, including limited infrastructure, ethical considerations, policy gaps, and resistance from educators (Funda & Piderit, 2024).

In Ghana, AI is increasingly used to tackle educational challenges and improve learning outcomes through innovative tools like AI-driven tutoring systems, chatbots, and practical-based teaching models. For example, Brilla AI supports science and mathematics education across Africa with interactive quizzes, engaging students and potentially broadening access to quality educational resources (Boateng et al., 2024). Generative AI further enhances cognitive skills by connecting theoretical concepts to real-world applications, offering holistic assessments beyond traditional exams (Nyaaba, 2023).

One notable tool, Rori, is an AI-powered math tutor available on WhatsApp in Ghana, which has significantly improved students' arithmetic skills. Rori exemplifies the positive impact of affordable, personalised learning tools in regions with limited educational resources. Additionally, research on KNUSTBot, an AI chatbot used in higher education, found that students who engaged with it outperformed their peers who relied solely on human instructors. The chatbot's instant feedback and 24/7 availability increased students' engagement with course content, fostering greater confidence and enhanced performance (Henkel et al., 2024).

In Kenya, AI is transforming education through innovative teacher-AI collaborations and digital personalised learning tools. For instance, AI helps teachers tailor content to meet the specific needs of pre-primary students, leading to more effective device usage and improved digital learning outcomes (Sun et al., 2024). Kenya is home to several impactful AI-driven platforms: M-Shule, a mobile platform, delivers personalised lessons to primary students via SMS, using performance data to adapt content to each learner's needs. iMlango combines AI with ed-tech



to monitor absenteeism and offers a multimedia learning platform that targets at-risk students through data analytics. Additionally, Maths-Whizz provides adaptive support for students struggling with math, showcasing AI's capacity to foster personalised learning. Kenya's advancements in AI are bolstered by supportive government policies and infrastructure, underscoring AI's potential in the educational sector. However, expanding AI's benefits across sectors requires increased public-private collaboration to further enhance infrastructure and accessibility (Ndungi, 2023).

The Nigerian Perspective

The current Artificial Intelligence (AI) environment in Nigeria's education sector combines great potential with significant obstacles. While the adoption of AI in Nigerian education is still in its early stages, it is gradually gaining positive traction among educators. University lecturers, particularly those at the University of Ibadan, recognise AI as a valuable tool for enhancing teaching, facilitating personalised learning, supporting research, and alleviating workloads. 87.1% of Nigerian lecturers expressed interest in additional AI training, signalling a strong willingness to adopt AI tools to improve education (Ezekiel & Akinyemi, 2022). According to research, Nigerian instructors have a moderate understanding of AI but greatly desire more discipline-specific training, notably in the arts (Bali, 2024). This demonstrates that, while educators are beginning to see the benefits of AI, a lack of practical knowledge and skills may be preventing widespread use. AI implementation varies significantly across academic areas. AI is more established in technical subjects like computer science, but its integration into the humanities and social sciences remains in its early stages.

Disparities in AI adoption in education arise from differences in perceived relevance and digital literacy among educators. In science education, teachers require training in AI tools such as adaptive learning systems, intelligent tutoring systems, and virtual labs to address shortages of qualified educators and provide hands-on learning. Training should enable teachers to use AI for personalised learning, data insights, and promoting scientific thinking (Okunade, 2024). In mathematics education, training on AI tools, including ChatGPT, should emphasise its potential and limitations, teaching educators how to integrate it into lesson plans to facilitate interactive learning and personalised feedback (Egara, 2024). AI integration necessitates an understanding of cultural context and translation nuances for the arts and humanities. Training should cultivate skills in using AI for creative exploration, research, and critical thinking, with an emphasis on ethics, including plagiarism and responsible AI use (Ibrahim, 2024). Training varies by education level. For K-12, training should encompass foundational AI concepts, age-appropriate tools, and strategies for adapting AI to diverse resources and student needs (Sanusi et al., 2024). In higher education, training should incorporate advanced AI topics, research integration, curriculum design, and AI ethics to prepare students for AI careers (Afolabi, 2024).

Despite limitations, AI is advancing Nigerian education through adaptive learning systems, intelligent tutoring systems, and virtual labs. Tools like uLesson are transforming engagement and outcomes. uLesson aligns with the Nigerian curriculum, offering video lessons, quizzes, and adaptive technology that tailors content based on student performance. Targeting secondary students preparing for WAEC and JAMB exams has positively affected comprehension, particularly in math and sciences. Some lessons are also available in local languages, thus expanding accessibility. Beyond adaptive learning, virtual laboratories present promising solutions to Nigeria's STEM education challenges. Due to difficulties in providing physical labs, the National Open University of Nigeria (NOUN), which has over 500,000 students across 110 centres, is exploring virtual labs. Currently, STEM labs are available in only thirteen locations, forcing students from other centres to travel, which leads to low lab attendance. NOUN views virtual labs as a means to ensure that all STEM students have equitable access to practical learning (Ndunagu et al., 2023). Virtual labs could benefit Nigeria by being cost-effective, flexible, and accessible. Unlike traditional labs, which are expensive to build and maintain, virtual labs reduce costs by eliminating the need for physical infrastructure, allowing institutions to reallocate resources. However, implementing virtual labs faces several challenges: limited infrastructure, unreliable internet, inconsistent power supply, and high expenses associated

with necessary software, hardware, and training. Many institutions lack the resources and technical expertise required to manage these platforms. A balanced model that combines virtual and physical labs could offer a more comprehensive learning experience, providing students with the accessibility of virtual labs alongside the hands-on experience of physical labs (Ndunagu et al., 2023).

Advanced AI technologies, such as intelligent tutoring systems and learnable robots, are less prevalent in Nigeria than in other nations (Bali, 2024). However, promising developments are underway, with several pilot projects exploring AI applications and government initiatives beginning to support AI's role in education (Okunade, 2024). Nigerian researchers are actively investigating various AI applications that suggest a broader future for AI in education. These include AI in e-learning platforms, evolutionary software modelling, student performance prediction, multimedia learning platforms, and the integration of AI with Moodle, a widely used learning management system (Bali, 2024). Together, these efforts represent a foundational but growing use of AI, backed by government initiatives and pilot projects aimed at expanding and enhancing AI's impact on Nigerian education.

Challenges Facing AI Utilization In Nigerian Education

This offers a detailed examination of the complexities surrounding the adoption of AI technologies in Nigeria's education sector. Infrastructure deficiencies in sub-Saharan Africa, particularly in Nigeria, present significant challenges to utilising AI in education. With only 18% of students having internet access and facing high data costs, many cannot engage with online resources. Computer access is also limited, with just 11% of students having a household computer. Many schools restrict personal devices, and frequent power outages hinder access to digital resources (Bali, 2024).

Geographical and socio-economic disparities further exacerbate the digital divide, particularly in rural areas where unreliable electricity limits access to AI education. Financial constraints prevent students from acquiring necessary devices, and gender disparities in technology access persist, with more male participants in educational programs (Ukeje et al., 2024).

Teacher preparedness is crucial for effective AI integration in classrooms. Educators must possess technical skills and understand pedagogical methods to use AI tools effectively. A study in Nigeria revealed that mathematics teachers lacked familiarity with tools like ChatGPT due to inadequate training and unclear integration guidelines (Onyejegbu, 2023). Essential skills include AI literacy, digital literacy, critical thinking, and effective assessment practices. Ethical considerations surrounding AI in education are pressing, with a lack of research on AI ethics in Nigeria. There are concerns about overreliance on AI, which may hinder students' critical thinking and creativity. Additionally, collecting extensive student data raises issues of data security and potential biases in AI assessments, impacting students from diverse backgrounds (Okunade, 2024).

Funding remains a major barrier to AI adoption in Nigerian education. Despite having reputable academic institutions, inadequate financial support limits the ability to implement AI initiatives effectively. Investment in infrastructure, hardware, software, localised content development, and ongoing teacher training is crucial. This situation highlights the need for government support, public-private partnerships, and strategic resource allocation to tackle financial challenges in AI education (Ogunode, 2020).

Areas of Further Study

Integrating AI in education presents numerous opportunities for advancing research in this field. Emerging areas include the development of AI-powered educational tools, teacher-AI collaboration model, government policy and support, research and development initiatives and the exploration of generative AI's impact on learning. These areas offer potential for creating more engaging, inclusive, and personalised educational experiences. The following sections outline key research directions in AI in education.



- 1. Development of AI-Powered Educational Tools:** Integrating AI into Nigeria's education system holds significant potential to address key challenges like limited tech access, low digital literacy, and connectivity disparities. Research should prioritize developing inclusive, scalable, and affordable AI-driven platforms tailored to Nigeria's unique educational needs, particularly in underserved areas. Key opportunities include personalized AI tutoring to adapt learning, 24/7 chatbot academic support, IoT attendance systems for accountability, and predictive analytics to identify at-risk students (drawing lessons from South Africa). Furthermore, adapting mobile education via SMS with AI-powered assessments and localized content is crucial for rural accessibility. Success hinges on collaboration between government, telecom providers, and educational stakeholders, alongside efforts to enhance digital literacy among both educators and parents. Ultimately, strategic AI integration can bridge educational gaps and improve learning outcomes across Nigeria.
- 2. Solar Energy Solutions:** Research should explore how solar energy can sustainably power AI integration in Nigeria's education sector by ensuring reliable electricity for broadband, data centres, and digital tools, especially given the country's power challenges. Feasibility and cost-effectiveness studies are needed for deploying solar systems in schools, particularly in underserved rural areas, to power internet, offline AI, and learning devices. Integration strategies with existing programs, emphasizing scalability and sustainability, are also crucial. Establishing solar-powered Digital Innovation Hubs can bridge the digital divide by offering digital literacy, AI education, coding, and ethical technology use, with research evaluating their community impact and role in digital inclusion. Furthermore, ethical considerations necessitate developing policies for data protection, algorithm transparency, and human oversight. Finally, research should assess the role of partnerships with local and international organizations in mobilizing resources and expertise for context-specific solar-powered AI education solutions.
- 3. Teacher-AI Collaboration Models:** Integrating AI into Nigerian education holds transformative potential but faces hurdles in teacher preparedness, access to resources, cultural acceptance, funding, and policy implementation. Research should prioritize strategies for embedding AI into teacher education curricula, equipping future educators to use AI for personalized learning, data-driven instruction, and ethical considerations. Furthermore, professional development frameworks are needed to support in-service teachers with continuous training and resources, addressing digital literacy gaps and varying technology access across Nigeria, potentially through mobile or offline solutions. Overcoming cultural resistance requires awareness campaigns, community engagement, and pilot projects showcasing AI's practical benefits in improving learning outcomes.
- 4. Government Policy and Support:** A critical research area is the development of a specific policy framework for AI-integrated EdTech in Nigeria. Current policies lack detailed guidance on the ethical, transparent, and inclusive application of AI in education. Research should identify policy gaps and propose updates advocating for government prioritization of technology in education through dedicated funding and improved infrastructure, especially in underserved areas. A national AI in education strategy is essential, outlining the integration of AI concepts across all educational levels and addressing resource allocation and teacher training. Furthermore, research must focus on the ethical implications, proposing policies that ensure transparency, unbiasedness, and equitable access to AI, addressing data privacy, algorithmic bias, and digital equity within Nigeria's diverse socio-economic and infrastructural context.
- 5. Collaborative Research Projects:** Collaborative research among universities, tech companies, government, and civil society is vital for developing tailored AI solutions for Nigeria's educational challenges, especially in underserved areas. Research should evaluate AI tool effectiveness across diverse classrooms and explore public-private partnerships and CSR for investment. A key focus should be creating offline AI tools and localized applications (speech/text in native languages) to overcome internet limitations and enhance inclusivity. Assistive AI technologies

for students with disabilities are also crucial. Furthermore, developing regulatory and ethical frameworks that promote investment while ensuring transparency, unbiasedness, and cultural sensitivity is essential.

6. Generative AI in Education: Integrating generative AI into Nigeria's education offers transformative potential for practical learning through simulations and interactive scenarios, addressing resource limitations. Research should focus on its adaptability for personalized learning across Nigeria's diverse student population, fostering creativity and critical thinking while aligning with cognitive and practical needs. Understanding generative AI's impact on educational roles and ensuring its ethical and equitable integration, complementing rather than replacing teachers and aligning with cultural goals, is crucial. Research must also address deployment challenges in resource-constrained environments by exploring affordable, scalable, and low-bandwidth solutions, alongside comprehensive teacher training for effective utilization.

Conclusion

Nigeria has a significant opportunity to transform its educational landscape by drawing inspiration from successful technological implementations across Africa. The proposed future work areas, including developing AI-powered educational tools, IoT-enabled attendance systems, and data-driven interventions, can enhance learning outcomes and address existing challenges such as dropout rates. Additionally, adapting mobile learning platforms and integrating practical-based teaching methods will foster a more engaging educational experience. Professional development for educators and the establishment of ethical AI practices is vital for ensuring that technology is used responsibly and effectively. Furthermore, community engagement initiatives and collaborative research projects can build a supportive ecosystem that empowers students, parents, and educators. To realise this vision, it is essential to advocate for robust government policies that provide the necessary funding and infrastructure improvements. By implementing these strategies, Nigeria can improve its educational outcomes and prepare its future workforce for the demands of a rapidly evolving digital economy. Embracing these advancements will help create a more equitable and innovative education system that benefits all Nigerians.



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About the Author

Mary Oluwagbemisola Adetula is a Doctoral Candidate at Obafemi Awolowo University in Nigeria, specialising in Political Science. Her expertise encompasses democratic governance, gender and development, and the dynamic interplay between ICT and the political economy. Mary has a particular interest in the transformative impact of AI on education, with a focus on its applications for teaching and learning in Nigeria, while drawing comparisons with AI integration across other African nations. Her contributions demonstrate a commitment to understanding how technology can shape political and social landscapes. She is based in Ile-Ife, Osun State, Nigeria.



From Promise to Progress: Making AI Work for Africa is a visionary collection that explores how AI can be a force for inclusive, sustainable development across the continent. This timely volume brings together leading voices to examine the current state of AI in Africa, its successes, its shortcomings, and its immense potential. Through regionally grounded chapters, the book provides a comprehensive examination of how AI is being applied in healthcare, agriculture, education, and community development. It highlights real-world innovations, from mobile health diagnostics to climate-smart farming, and outlines bold strategies for overcoming infrastructure, policy, and ethical challenges. At its heart, this book champions a people-first approach to AI, one that centres African values, amplifies community voices, and ensures that technological progress translates into meaningful social impact. Whether you're a researcher, policymaker, technologist, or development practitioner, *From Promise to Progress* is your essential guide to building an AI-powered future that works for all Africans.

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